BYO AI ACCOUNT BRIDGE / RESEARCH SERIES

Product Requirements

Personas, user journeys, scope, UX states and acceptance standards

02

DETAILED TECHNICAL DOSSIER

User-owned accounts / Local-first runtime / Optional cloud relay

Research cutoff: 26 October 2026

A Consumer-web adapter (permission-gated)

B Official account authorization where supported

C User-owned free APIs and locally running models

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> Research notation: [Sxx] markers link to first-party terms, official technical documentation, published standards or original project repositories. Statements labeled proposed, target, or estimate are design judgments, not observed performance.

Product charter and user promise

Working product label: BYO AI Bridge. This is a functional project name, not a trademark clearance result. **North star:** A developer with an existing account or permitted free model can turn that authorized access into a usable, transparent API-like endpoint without becoming an expert in model hosting or gateway configuration.

Product boundary: User-owned connections only. The desktop agent is the trusted execution boundary for accounts and inference. The website is the management plane and, only when opted into, a remote request relay. A provider's own contract determines whether an A-style consumer session can be automated. **[S01] [S02] [S03]**

> POSITIONING: "Connect the AI access you already own. Build with one local endpoint. Know which model answered, which account paid or supplied the quota, and what happens when it runs out."

Explicit non-promises

- No unlimited prompts or guaranteed access to proprietary premium models.
- No cross-user account pooling, shared subscriptions or secret scraping at the provider login wall.
- No claim that a web-generated response implements every field or feature of an actual model API.
- No assurance that an online dashboard can serve requests while the user's bridge is offline.
- No automatic spending on a paid route unless the user has opted in with a hard budget.

Target users, jobs and use cases

Persona	Current frustration	Job to be done	First-use moment
Budget-limited student	Cannot justify pay-as-you-go API spending	Prototype a course project with available access	Connect a free API or permitted account and test curl
Indie developer	Several models/accounts with different formats	Switch models without changing app code	OpenAI SDK example works against localhost
Coding-agent power user	Tool only accepts an OpenAI-compatible base URL	Use approved account or free model in existing tool	Configure one endpoint and model ID
Privacy-sensitive hobbyist	Does not want account cookies held by a startup	Execute locally and control remote access	See Local-only mode and local credential indication
Small team (later)	Different quotas, auditing and ownership	Centralized policy without sharing personal consumer sessions	Not in initial MVP

Primary user: budget-limited individual developer running a laptop or desktop. **Secondary user:** advanced tool user who values local account isolation. Do not optimize v1 for production SaaS companies that require 24/7 uptime, strong SLAs or regulated-data workflows.

Core user journeys

Journey J01 - First local response

- User installs the signed local bridge and opens its control page.
- Onboarding explains the difference among web (A), account authorization (B), and official API/local (C).
- User chooses one eligible connection; unsupported connectors are labeled honestly.
- Any login is completed in a provider-controlled flow or a local browser session; the dashboard never asks the user to paste account cookies.
- Connection appears with status, provider, capability class and quota caveat.
- User copies a local base URL and user-generated gateway key.
- A built-in test sends a short user prompt; a completed normalized response appears with timing and source label.

Acceptance: For an eligible user with working internet and available quota, target under five minutes to first supported text response. Measure this in pilot; it is not an existing performance claim.

Journey J02 - Remote access from another device

- User signs into the dashboard and enrolls one bridge device using a short-lived pairing code.
- The local app shows the destination platform, device identity and exact data-visibility warning.
- User explicitly enables remote relay. Agent establishes outbound authenticated TLS/WebSocket.
- User creates a scoped remote gateway key and specifies a target device/profile.
- Remote client calls the gateway. Dashboard authorizes the key, finds a live device and forwards the request.
- Local bridge chooses the authorized connector; response streams back through relay.
- If the agent disconnects, the request fails with `device_offline` instead of switching to another user's account.

Acceptance: No inbound home-router port required; remote access can be disabled and revoked from both device and dashboard. WebSocket identity and per-request authorization follow security guidance. **[S19]**

Journey J03 - A quota is exhausted

- Provider returns or exposes a trustworthy limit status.
- Gateway classifies `quota_exhausted`, shows the provider and known retry/reset information, and stops repeat requests to that connector.
- If the user pre-authorized a compatible free or local fallback, the router may offer it, respecting privacy and capability settings.
- If no approved fallback exists, the client receives a documented error and the dashboard displays a human explanation.
- Paid fallback remains disabled by default; no hidden use of account credits. **[S07] [S33]**

Journey J04 - Revoke or disconnect

User disconnects a provider and the local agent removes its reference; for official OAuth, revoke where supported and clear local credentials; for web profiles, allow sign-out and local profile deletion after confirmation. Cloud retains at most non-sensitive connection metadata until its retention period. OpenAI's grant lifecycle requires token refresh and correct handling of revoked grants. **[S32] [S33]**

Functional requirements with priority

ID	Requirement	Priority / measurable condition
FR-01	Normalized `/v1/chat/completions` text response	P0; schema-valid result with truthful source metadata
FR-02	`GET /v1/models` based on actual connection capabilities	P0; no unavailable model is advertised as working
FR-03	Local API key protection even on loopback	P0; no unauthenticated generation
FR-04	A adapter SDK with signed-in, degraded, denied states	P0; mock browser adapter tested end-to-end
FR-05	B adapter for approved account-plan inference	P1; only when user is eligible, supported model/route **[S04]**
FR-06	C adapter for one free API and local Ollama	P0; fallback explicitly configured **[S09]** **[S13]**
FR-07	Quota and error normalization	P0; distinguish quota from invalid credentials
FR-08	Streaming for truly streaming-capable adapters	P1; completed terminal event required
FR-09	Dashboard device enrollment and status	P1; online/last-seen status correctly reported
FR-10	Remote relay with explicit opt-in	P1; cannot invoke disabled/offline device
FR-11	Per-user and per-device scoped gateway keys	P1; revocation effective within documented window
FR-12	MCP tool facade	P2; `ask_model` routes through same policy checks
FR-13	Advanced tools/vision/JSON strict schema	P2+; only after per-provider capability verification
FR-14	Account deletion and export	P1; delete cloud account and associated metadata

UX information architecture

Local app navigation

View	Content	Primary action
Home	Bridge status, local URL, one-call test	Copy connection snippet
Connections	A/B/C categories, health, permissions, capability badges	Connect an available provider
Models	Available model IDs, features and known limitations	Choose preferred route
Routing	Ordered eligible fallback list with explicit consent	Save routing policy
Activity	Request count, latency, status, privacy-safe errors	Inspect local diagnostics
Settings	API key, startup, data retention, remote relay	Turn relay on/off

Online console navigation

View	Content	Primary action
Dashboard	Devices online, request totals, last errors	Open device
Devices	Name, ID, enrollment status, connectivity	Enroll or revoke
Keys	Last four characters, scopes, device binding	Rotate key
Remote API	Endpoint, example, status and privacy warning	Enable per-device relay
Usage	Per-user metadata, no prompt text by default	Export metrics
Account	Privacy choices, deletion, sessions	Revoke and delete

UX states and expected behavior

State	Message to user	API behavior
`ready`	Connected and available for compatible requests	Serve normally
`needs_login`	Sign in again to this account	401 + `login_required`
`quota_exhausted`	Usage limit reached; reset may be unknown	429 + retry hint if known
`unsupported_feature`	This connection cannot use tools/files/etc.	400 with capability detail
`provider_unavailable`	Provider temporarily unavailable	503; no infinite retries
`policy_disabled`	Connector unavailable until authorization is confirmed	403, not workaround advice
`device_offline`	Start your computer's bridge to use remote API	503 with device ID
`remote_disabled`	Remote access is off by your choice	403, no tunnel negotiation

Honest capability copy examples: "Text available; JSON mode is best effort"; "Model chosen by provider UI; exact model unverified"; "Account quota is managed by the original provider"; "Remote mode routes prompt and output through the dashboard relay"; "Current API route cannot access file search or image generation." **[S05]**

Design principles and accessibility

- **Never obscure the source.** Every response and activity row identifies connection class, provider and model confidence (`verified`, `provider-declared`, or `unknown`).
- **Minimal secret handling.** No web form asks users to paste session cookies; OAuth opens the provider's own authorization page.
- **Legible state.** Color is never the only indicator of connection health; use text and accessible icons.
- **Keyboard usability.** Every connect, disconnect, copy and consent action is reachable without a pointer; errors are announced to assistive technologies.
- **Privacy controls near the choice.** The remote enable switch includes disclosure before activation, not hidden in long terms.
- **Explicit consumption.** Display "included plan allowance", "free API quota", "local compute" or "paid fallback" as applicable.
- **Safe defaults.** Local-only mode, no prompt logging, no paid routes, conservative concurrency, and clear data deletion controls.

Success metrics and research instrumentation

Metric	Definition	Pilot decision threshold (target)
Time to first response	Install start to first successful compatible response	Median under 5 minutes
Local success rate	Completed valid simple text requests / eligible requests under quota	At least 95% in controlled test
Accurate errors	Known faults returning correct normalized error class	100% of scripted fault cases
Account isolation	Cross-user / cross-device request failures	Zero unauthorized successes
Week-two retention	Pilot users active in week two / activated users	Learn baseline; no invented forecast
User trust	Interview comprehension of web vs OAuth vs free API	90% correctly explain their connection

Collect event counts and durations rather than raw prompts. Define event schema `request_completed`, `request_failed`, `provider_connected`, `permission_changed`, and `device_offline`. Use stable pseudonymous identifiers and permit local analytics opt-out. Do not interpret free-tier quotas as reliable capacity guarantees. **[S10] [S11]**

Acceptance gates and product exclusions

Release a connection only after: functional test, documented permission assessment, capability truthfulness, credential isolation test, rollback mechanism and a clear user-facing limitation. A connector that merely works through an unofficial browser session does not pass the permission gate.

Out of scope for v1: mobile apps, team account pooling, 24/7 hosted inference, automatic creation of provider accounts, artificial limit circumvention, full OpenAI/Anthropic compatibility, cross-device chat history synchronization, web crawling and enterprise compliance certifications.

PRD decision log

Decision	Status	Why
Local agent as execution authority	Recommended	Keeps account secrets local and supports offline local use
Web adapter first-class but permission-gated	Required	Preserves the central product idea without an unapproved shipping claim
API before MCP	Recommended	Existing apps and SDKs can call ordinary REST; MCP exposes tools **[S14]**
Cloud metadata only by default	Required	Limits credential and sensitive content exposure
Remote relay optional	Required	The cloud path has additional trust and availability costs
Open-source local core	Proposed	Fits the affordability mission and permits external review

Evidence & source register

Sources were checked in October 2026. Their content, access conditions, prices and quotas can change. Project READMEs are evidence of what authors claim to implement, not proof of provider permission or reliability.

[S01] OpenAI consumer Terms of Use (effective January 2026)
https://openai.com/policies/row-terms-of-use/

[S02] Anthropic Consumer Terms of Service (effective October 2025)
https://cdn.sanity.io/files/4zrzovbb/website/43c13f51a6d9755341433f79cf63c966ed9200ea.pdf

[S03] Anthropic Claude Code - Legal and compliance
https://code.claude.com/docs/en/legal-and-compliance

[S04] OpenAI Sign in with ChatGPT - Local open-source plan usage
https://developers.openai.com/siwc/token-sharing-open-source

[S05] OpenAI Sign in with ChatGPT - Preview limitations
https://developers.openai.com/siwc/token-sharing-open-source/preview-limitations

[S07] OpenAI Help - Using ChatGPT plans in other apps
https://help.openai.com/en/articles/20001542-using-your-chatgpt-plan-in-other-apps-and-sites

[S09] Google Gemini API billing and free tier
https://ai.google.dev/gemini-api/docs/billing

[S10] Google Gemini API rate limits
https://ai.google.dev/gemini-api/docs/rate-limits

[S11] GroqCloud API rate limits
https://console.groq.com/docs/rate-limits

[S13] Ollama OpenAI-compatible API
https://docs.ollama.com/api/openai-compatibility

[S14] MCP specification release, 28 July 2026
https://blog.modelcontextprotocol.io/posts/2026-07-28/

[S19] OWASP WebSocket Security Cheat Sheet
https://cheatsheetseries.owasp.org/cheatsheets/WebSocket_Security_Cheat_Sheet.html

[S32] OpenAI - Sign in with ChatGPT account authorization
https://developers.openai.com/siwc/token-sharing-open-source/sign-in

[S33] OpenAI - Sign in with ChatGPT errors and recovery
https://developers.openai.com/siwc/token-sharing-open-source/errors-and-recovery

> Evidence boundary: this series is a design-and-research package, not a working implementation, legal opinion, measured security certification or promise of free access to any specific proprietary model.